JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Logarithm

JEE Main Mathematics · Previous Year Questions with Answer Key

11	11	2020 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Numerical	6 questions
MCQ (Single Correct Answer)	5 questions
Answer key	at the end

1 **Solve**
Print it, or work through it on screen.

2 **Check**
The answer key is on the last pages.

Numerical 6 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

The sum of squares of all the real solutions of the equation

$\log_{(x+1)}\left(2x^2+5x+3\right)=4-\log_{(2x+3)}\left(x^2+2x+1\right)$ is equal to ____ .

Q2. JEE Main 2023 (Online) 10th April Morning Shift

Let a, b, c be three distinct positive real numbers such that $(2a)^{\log_e a}=(bc)^{\log_e b}$ and $b^{\log_e 2}=a^{\log_e c}$.

Then, 6a + 5bc is equal to _____.

Q3. JEE Main 2023 (Online) 25th January Morning Shift

Let $S=\left\{\alpha:\log_2(9^{2\alpha-4}+13)-\log_2\left(\frac{5}{2}.\ 3^{2\alpha-4}+1\right)=2\right\}$. Then the maximum value of β for which the equation $x^2-2\left(\sum_{\alpha\in S}\alpha\right)^2 x+\sum_{\alpha\in S}(\alpha+1)^2\beta=0$ has real roots, is _____.

Q4. JEE Main 2021 (Online) 20th July Evening Shift

The number of solutions of the equation

$\log_{(x+1)}(2x^2+7x+5)+\log_{(2x+5)}(x+1)^2-4=0$, x > 0, is :

Q5. JEE Main 2021 (Online) 26th February Morning Shift

The number of solutions of the equation $\log_4(x-1)=\log_2(x-3)$ is _____.

Q6. JEE Main 2020 (Online) 9th January Morning Slot

The number of distinct solutions of the equation
$\log_{\frac{1}{2}}|\sin x|=2-\log_{\frac{1}{2}}|\cos x|$ in the interval [0, 2π], is ____.

MCQ (Single Correct Answer) 5 questions

Q7. JEE Main 2026 (Online) 4th April Evening Shift

Let $\alpha = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots \infty$ and

$\beta = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots \infty$. Then the value of

$(0.2)^{\log_{\sqrt{5}}(\alpha)} + (0.04)^{\log_5(\beta)}$ is equal to :

(A) 4 (B) 5 (C) 8 (D) 25

Q8. JEE Main 2026 (Online) 23rd January Evening Shift

The sum of all the real solutions of the equation
$\log_{(x+3)}\left(6x^2+28x+30\right) = 5 - 2\log_{(6x+10)}\left(x^2+6x+9\right)$ is equal to :

(A) 1 (B) 4 (C) 0 (D) 2

Q9. JEE Main 2025 (Online) 22nd January Morning Shift

The product of all solutions of the equation $e^{5(\log_e x)^2+3} = x^8, x > 0$, is :

(A) e^2 (B) e (C) $e^{6/5}$ (D) $e^{8/5}$

Q10. JEE Main 2023 (Online) 11th April Morning Shift

The number of integral solutions x of $\log_{(x+\frac{7}{2})}\left(\frac{x-7}{2x-3}\right)^2 \geq 0$ is :

(A) 8 (B) 7 (C) 5 (D) 6

Q11. JEE Main 2023 (Online) 30th January Morning Shift

If the solution of the equation $\log_{\cos x}\cot x + 4\log_{\sin x}\tan x = 1, x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha+\sqrt{\beta}}{2}\right)$, where α, β are integers, then $\alpha+\beta$ is equal to :

(A) 3 (B) 6 (C) 4 (D) 5

Answer Key

Numerical

1. **2**	2. **8**	3. **25**	4. **1**	5. **1**
6. **8**				

MCQ (Single Correct Answer)

7. **C**	8. **C**	9. **D**	10. **D**	11. **C**